

ASSESSMENT 3 BRIEF	
Subject Code and Title	DLE602 Deep Learning
Assessment	Deep Learning Final Project
Individual/Group	Group
Length	1,500 words (+/–10%) Report and Source Code
Learning Outcomes	The Subject Learning Outcomes demonstrated by the successful completion of the task below include: c) Develop critical analysis skills in deep learning research. d) Demonstrate collaborative skills to apply deep learning to solve real world problems. e) Demonstrate ability to effectively communicate scientific and technical information.
Submission	6-week delivery: Due by 11:55pm AEST/AEDT, Wednesday of Week 6 12-week delivery: Due by 11:55pm AEST/AEDT, Wednesday of Week 12
Weighting	40%
Total Marks	100 marks

Task Summary

In Assessment 2, your group reviewed quality research articles in the field of deep learning and proposed a research project for implementation based on the literature review. You also presented your project proposal to wider audience in a 5–7-minute audio-visual seminar.

In Assessment 3, you and your group members will continue to work on the proposed project and will implement the project based on the Deep Learning principles discussed and learned in this subject.

Your group will also produce a professional report detailing the entire implementation process, including a clear list of the requirements/functionalities, the steps taken to address those requirements, the deep learning principles considered for the implementation, the methods used and an analysis of the outcomes.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Context

The aim of this assessment is to gain implementation experience of a deep learning project concerning a real-life issue. The written report should demonstrate your ability to practically implement the theories you learned in the modules.

Task Instructions

Your group will implement the project identified in the previous assessment using Python programming language. The code must be well formatted and conform to Python naming conventions. You also need to provide sufficient comments in the code.

You are also required to prepare a 1500-word report in which you will integrate knowledge from your literature review and your Assessment 2 report. You need to highlight all the key points that you considered for implementation and demonstrate what you have undertaken to complete the key points. In the report, you will also discuss the process of implementation and the outcomes.

The report should demonstrate that the group fully understands the deep learning concepts learned in the subject. The report should provide concluding remarks that show the analysis and synthesis of ideas.

It should be noted that this is a **new report** and not the same as that produced in Assessment 2. In the previous report, you defined the project. In this assessment, your focus should be on the implementation of the project and demonstrating your knowledge of deep learning.

Report Guidelines

In relation to the report, you need to:

- Prepare a 1,500-word report in which you clearly articulate the entire process you adopted to implement your project and your project outcomes;
- Provide a concise summary of the previous assessment, including the project statement, aim and research question(s);
- Present your report using any standard writing format;
- Include a cover page and table of contents. Note: the cover page needs to include official student names and numbers;
- State the word count at the end of the report (before the reference section);
- Insert page numbers (the page numbers should appear in the footer of each page with your group id); and
- Label any figures and tables in your report with meaningful captions.

Referencing

It is essential that you use APA style to cite and reference your research. For more information on referencing, visit our Library website at: <https://library.torrens.edu.au/academicskills/apa/tool>.

Submission Instructions

Compile your written PDF or word report and your source code with instructions on how to execute the code in a zip file. Ensure that you have the names and student ID of the members of the group on the front page. Submit the zip file using Assessment link in the main navigation menu for 'DLE602 Deep Learning Assessment 3'.

Academic Integrity Declaration

We declare that except where we have referenced, the work we are submitting for this assessment task is our own work. We have read and are aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

We are aware that we need to keep a copy of all submitted material and their drafts, and we will do so accordingly.

Assessment Rubric

Assessment Attributes	Fail (Yet to achieve minimum standard) 0–49%	Pass (Functional) 50–64%	Credit (Proficient) 65– 74%	Distinction (Advanced) 75– 84%	High Distinction (Exceptional) 85– 100%
<i>Completeness and efficiency</i> <i>The implementation covers all requirements</i> <i>The whole system is easy to use and run</i> Percentage for this criterion = 20%	None of the requirements have been implemented. The system does not function properly or is extremely buggy. An extreme level of manual configuration is required to run the system. Additionally, the configuration does not work.	One or two major requirements have been implemented. The system does not function properly. No exception handling has been implemented. Users are required to follow a lengthy configuration manual to run the system.	All but one or two major requirements have been implemented. The system functions only if certain additional conditions are met. Basic exception handling has been implemented, but it is not thorough. Users are required to follow a short configuration manual to run the system.	Most of the major requirements have been implemented. The system functions without any additional conditions needing to be met. Basic exception handling has been implemented, but it is not thorough. Users are only required to copy the necessary data in the right locations.	All of the major requirements have been implemented. The system functions properly. Exceptions are handled very well. Users can run the system without any configuration.
<i>Coding Convention and Quality of Code</i> <i>The code follows a consistent naming convention</i> <i>The code is well formatted and includes appropriate spacing and indentation</i>	The code is not formatted. Little or no comments are provided. The naming of the methods or variables is inconsistent. No naming convention is followed.	There are significant errors in the format of the code and the naming of the methods or variables. There is a significant lack of useful comments.	The code is generally well written, but there is some room for improvement. There are more than five errors but less than eight errors in terms of the naming conventions and the format of the code.	The code is generally well written. There are more than two errors but less than five errors in terms of the naming conventions and the format of the code.	The code is expertly written. There are no more than two errors in terms of the naming conventions and the format of the code.

<i>The code contains sufficient comments; the implementation of the major functions are commented upon</i> Percentage for this criterion = 20%			There is a reasonable amount of useful comments.	There is a sufficient amount of useful comments.	There is a sufficient amount of useful comments.
<i>Integration of Knowledge, Topic Focus, Depth of Discussion</i> <i>Clear requirements</i> <i>Articulation of deep learning principles and necessary methods</i> Percentage for this criterion = 30%	The list of requirements/functionalities and project scope are not in sync. The review of the implementation, deep learning principles and methods does not highlight the relationship between the theories and the implementation. No analysis of the outcomes is presented.	The list of requirements/functionalities and project scope barely match. The review of the implementation, deep learning principles and methods barely highlight the relationship between the theories and the implementation. A short analysis of the outcomes is presented.	The list of requirements/functionalities overfits or underfits the project scope. The review of the implementation, deep learning principles and methods loosely highlight the relationship between the theories and the implementation. A high-level analysis of the outcomes is presented.	The list of requirements/functionalities manageable within the project scope is somewhat clear. The review of the implementation, deep learning principles and methods highlight the relationship between the theories and the implementation. A critical analysis of the outcomes is presented.	Presents a clear list of the requirements/functionalities manageable within the project scope. Excellent review of implementation, deep learning principles and methods. A clear relationship between the theories and implementation is established. A comprehensive and critical analysis of the outcomes is presented.

<i>Effective Communication (Written)</i> • <i>Writing skills</i> • <i>Organisation and structure</i> Percentage for this criterion = 15%	Poor writing skills. Additionally, the articulations are not clear at all. Lack overall organisation Very difficult to follow Grammar and spelling errors make it difficult for the reader to interpret the text in many places.	Writing is readable; however, it is difficult to comprehend the information presented. Not well organised for the most part. Difficult to follow. Choice of words needs to be improved. Grammatical errors impede the flow of communication.	Writing is readable and it is reasonably easy to comprehend the information presented. Organised for the most part. Difficult to follow. Words are well chosen; however, some minor improvements are needed. Sentences are mostly grammatically correct and contain few spelling errors.	Writing is good and it is easy to comprehend the information presented. Well organised. Cohesive and easy to follow. Words are well chosen. Sentences are grammatically correct and free of spelling errors.	Writing is excellent, short, sharp and to-the-point and easily digestible. Exceptionally organised. Highly cohesive and easy to follow. Words are carefully chosen that precisely express the intended meaning and support reader comprehension.
<i>Documentation</i> • <i>Headings and sub-topics formatting</i> • <i>Professional presentation</i> • <i>Referencing</i> Percentage for this criterion = 15%	Headings are used and not numbered. They are not clear in meaning. Formatting appears to be insufficient. Most of the tables and figures have been labelled but contain errors.	Headings are used (preferably numbered). Formatting appears to be sufficient. Tables and figures have been labelled with few errors.	Headings are clear in meaning (preferably numbered). Report is well formatted. Tables and figures are well labelled (where applicable). Sources are mostly cited professionally using the appropriate style guide (APA).	Headings are numbered and relate well to the discussions in the section. Tables and figures are labelled properly (where applicable). Sources are cited professionally, with a couple of exceptions, using the appropriate style guide (APA).	Report is professionally formatted. Headings follow a standard convention. Tables and figures are used and labelled properly. Sources are cited professionally using the appropriate style guide (APA).

The following Subject Learning Outcomes are addressed in this assessment

SLO c)	Develop critical analysis skills in deep learning research.
SLO d)	Demonstrate collaborative skills to apply deep learning to solve real world problems.
SLO e)	Demonstrate ability to effectively communicate scientific and technical information.